A STUDY ON

"IMAGE STEGNOGRAPHY USING REVERSIBLE LOGIC GATES"

In partial fulfillment of the requirements for the completion of the Degree of

BACHELOR OF TECHNOLOGY

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2025 – 2026

CERTIFICATE

This is to certify that the study entitled on **"IMAGE STEGNOGRAPHY USING REVERSIBLE LOGIC GATES"** Bonafide work done and submitted for the partial fulfillment of the requirements for the completion of B.Tech Degree in **ELECTRONICS AND COMMUNICATION ENGINEERING**, RGUKT RK Valley.

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DECLARATION

I hereby declare that the study entitled **"IMAGE STEGNOGRAPHY USING REVERSIBLE LOGIC GATES"** submitted to the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING** in partial fulfilment of requirements for the completion of the degree of **BACHELOR OF TECHNOLOGY.** This study is the result of my effort and that it has not been submitted to any other University or institution for the award of any Degree or Diploma other than specified above.

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TABLE OF THE CONTENTS:

S.NO	TITLE OF THE CONTENTS	PAGE NO
1	Title Page	1
2	Certificate	2
3	Declaration	3
4	Acknowledgement	4
5	Table of the contents	5
6	Introduction	6-8
7	Abstract	8-12
8	Existing model	13-15
9	Problem and Solution of the problem	15-16
10	Program code	17-21
11	Tools used	22
12	Simunlation results	23
13	Comparison table	24-25
14	Advantages and Disadvantages	25-27
15	Conclusion	28
16	References	28

INTRODUCTION

In the modern era of digital communication, enormous amounts of sensitive information are exchanged across public networks every second. As cyber threats, data interception, and unauthorized access continue to rise, ensuring secure data transmission has become a critical challenge. Traditional security techniques such as cryptography protect data by converting it into an unreadable form. However, encrypted data still attracts attention because it clearly signals that something confidential is being transmitted. This visibility itself can become a security risk. To overcome this limitation, the focus has shifted toward methods that conceal not just the content, but the very existence of information.

This leads to **steganography**, a technique that hides secret data inside a seemingly harmless medium such as an image, audio file, or video. A casual observer sees only a normal file, while hidden information remains completely invisible. Among various media, digital images are highly suitable for steganography because of their large pixel capacity and tolerance to minor modifications that remain undetectable to the human eye. By subtly embedding information into the least significant bits (LSBs) of pixel values, it becomes possible to conceal data without altering the visible appearance of the image. Thus, steganography enables covert and inconspicuous communication, making it a powerful tool in defense, medical confidentiality, digital watermarking, and secure data transfer.

However, most conventional steganographic systems are based on irreversible logic operations that inherently lose information during computation. According to **Landauer's principle**, every bit of information lost results in a minimum energy dissipation of $kT\ln\ 2$. As modern electronic devices continue to shrink into nano-scale dimensions, power dissipation and heat generation have become major bottlenecks in system performance and reliability. Irreversible logic consumes more power, generates more heat, and becomes less efficient as technology moves toward ultra-dense VLSI circuits and futuristic nano-computing platforms.

To address these limitations, researchers have turned toward **reversible logic**, a computing paradigm in which every output uniquely maps back to its input, ensuring that no information is

destroyed. Reversible circuits theoretically dissipate near-zero energy because they avoid information loss at the logical level. Among the fundamental reversible logic elements, the **Feynman (Controlled-NOT) gate** plays a crucial role. It performs copying and XOR operations without losing information, making it especially suitable for data manipulation tasks such as bit embedding and extraction in steganography. The reversible nature of the Feynman gate ensures that hidden information can be perfectly recovered, which aligns with one of the most important requirements of steganography—**lossless retrieval** of secret data.

In this project, the principles of reversible logic are applied to image steganography through a **Python-based implementation** that mimics Feynman gate behavior using XOR operations. A grayscale cover image and a secret image are processed pixel-by-pixel, where the two most significant bits of each secret pixel are embedded into the least significant bits of the cover pixel. The reversible XOR-based embedding ensures that the original secret data can be reconstructed perfectly during extraction, provided the stego image and cover image are available. The Python environment, supported by libraries such as PIL, NumPy, and scikit-image, enables efficient pixel manipulation and evaluation using quality metrics **like Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR)**, and **Structural Similarity Index (SSIM)**. High PSNR and SSIM values demonstrate that the stego image remains visually indistinguishable from the original cover, ensuring imperceptibility and security.

Beyond software implementation, this project carries significant importance in the context of **future hardware realization** using **Quantum Dot Cellular Automata (QCA)** technology. QCA is an emerging nano-electronic paradigm that replaces conventional current-driven CMOS transistors with electron position–based computing. Instead of voltage levels and current flow, QCA uses the arrangement of electrons within tiny quantum dots to represent binary logic. Neighboring QCA cells interact through electrostatic forces, enabling information propagation with **extremely low power consumption** and without conventional current switching. Importantly, several reversible gates—including the Feynman gate—have already been modeled and simulated using QCA structures, demonstrating that reversible logic is highly compatible with nano-scale QCA design.

This creates a strong conceptual bridge between the reversible embedding logic used in our project and potential future hardware implementations. Although our current system is purely software-based, the reversible XOR mechanism aligns closely with the behavior of QCA-based Feynman gate layouts. This suggests that the same embedding and extraction process could one day be implemented in physical QCA hardware, enabling ultra-low-power, nano-scale, real-time steganographic processing for applications in defense, secure IoT, biomedical systems, and quantum-inspired communication devices.

Overall, this project unifies three powerful domains—**steganography**, **reversible logic**, and **Quantum Dot Cellular Automata**—to demonstrate a secure, lossless, and energy-efficient approach to hidden communication. By proving the concept through Python simulation and aligning the logic with QCA principles, the work not only achieves effective data hiding but also lays a foundation for futuristic hardware systems capable of ultra-low-power secure processing at the nano scale. This makes the project highly relevant for the next generation of secure and intelligent communication technologies.

ABSTRACT

The rapid growth of digital communication has increased the need for secure and covert data transmission methods. Steganography offers a unique advantage by hiding the existence of information rather than merely encrypting it. This project presents an image steganography system inspired by reversible logic, specifically the behavior of the Feynman (CNOT) gate, to enable lossless embedding and extraction of secret data. A grayscale cover image and a secret image are processed pixel-by-pixel, where the two most significant bits of each secret pixel are embedded into the least significant bits of the cover pixel using a reversible XOR-based mapping. The system is implemented in Python using VS Code, and the extraction process applies the same reversible logic to perfectly reconstruct the hidden image. Quality metrics such as MSE, PSNR, and SSIM confirm that the stego image remains visually indistinguishable from the original, ensuring high imperceptibility and data integrity.

Beyond software validation, the project is conceptually aligned with Quantum Dot Cellular Automata (QCA), an emerging nano-electronic technology that enables ultra-low-power computation using electron positioning rather than current flow. Since reversible gates like the Feynman gate can be realized through QCA structures, the proposed embedding logic holds strong potential for future hardware implementation at the nano scale, offering reduced power dissipation and minimal heat generation compared to conventional CMOS-based systems. Thus, this work not only demonstrates an effective reversible steganography approach in software but also establishes a foundation for next-generation, energy-efficient, and quantum-inspired secure communication systems.

What is Steganography?

Steganography is the science and art of hiding secret information inside another ordinary, non-suspicious medium in such a way that the presence of the hidden data remains completely unnoticed. Unlike cryptography—which scrambles data into an unreadable form but still reveals that a secret message exists—steganography conceals the very existence of the information. A normal observer looking at the carrier file should not be able to detect that anything unusual is embedded inside it.

In digital systems, the most commonly used cover media include images, audio, video, and text files. Among these, images are particularly suitable because they contain a large number of pixels, allowing small modifications to be made without visibly changing the overall appearance. A popular method is **Least Significant Bit (LSB) embedding**, where the smallest bit positions of pixel values are altered to store secret data. Since changes in these lower bits cause minimal visual difference, the resulting image (called a *stego image*) appears almost identical to the original cover image.

The primary goal of steganography is invisible communication—to deliver a message without attracting attention or suspicion. It is widely used in secure communication, copyright protection, digital watermarking, medical data confidentiality, defense messaging, and forensic

applications. A successful steganographic system must ensure three key properties: imperceptibility (the stego image should look natural), security (embedded data must remain hidden from attackers), and recoverability (the hidden data should be accurately extracted by the intended receiver). Thus, steganography plays a vital role in modern information security by offering a covert layer of protection in digital communication.

What is Reversible Logic Gate?

Reversible logic refers to a computing model in which every output state uniquely maps back to its corresponding input state. In simpler terms, a reversible logic gate is one where the **number of inputs equals the number of outputs**, and no information is lost during computation. Because the original inputs can be perfectly reconstructed from the outputs, the process is logically reversible.

The importance of reversibility comes from Landauer's principle, which states that every bit of information lost in an irreversible operation dissipates a minimum amount of energy as heat. Traditional logic gates like AND, OR, and NAND are irreversible because their outputs do not uniquely define their inputs. As technology moves toward nano-scale VLSI, quantum computing, and ultra-low-power devices, such heat dissipation becomes a major challenge. Reversible logic addresses this by theoretically enabling computation with near-zero power loss, since no information is destroyed.

Reversible gates, such as the **Feynman (CNOT) gate**, **Toffoli gate**, and **Fredkin gate**, form the building blocks of reversible and quantum circuits. The Feynman gate, in particular, performs copying and XOR operations without information loss, making it valuable for data manipulation, signal fan-out, and bit transformation tasks. Because reversible logic preserves information and reduces energy dissipation, it is considered a foundational technology for future quantum processors, low-power VLSI architectures, cryptographic hardware, and secure data processing systems. In applications like steganography, reversibility ensures accurate recovery of embedded data, aligning perfectly with the need for lossless information extraction.

FEYNMAN GATE (CNOT GATE)

Definition

The **Feynman gate** is a 2×2 reversible gate with:

- Inputs: (A, B)
- Outputs:
  - P = A
  - Q = A ⊕ B

Because the outputs can reconstruct the inputs, no information is lost.

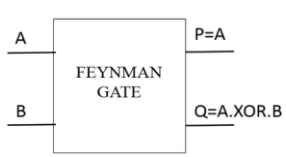
Functional Behavior

- If B = 0 → Q = A ⊕ 0 = A → acts as a **fan-out / copy** gate.
- If B = 1 → Q = A ⊕ 1 = ¬A → acts as a controlled NOT operation.

Role in Steganography

In this project, the XOR behavior of the Feynman gate is used to:

- Combine cover pixel bits and secret pixel bits reversibly.
- Recover the secret bits using the same XOR (Feynman-like) operation.



(a)

A	B	P	Q
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1

(b)

Quantum Dot Cellular Automata (QCA)

Quantum Dot Cellular Automata (QCA) is an emerging nanotechnology that offers a completely different way of implementing digital logic compared to conventional CMOS transistors. Instead of using current flow and voltage levels to represent logic '0' and '1', QCA uses the **position of electrons inside tiny quantum dots** within a cell. A basic QCA cell consists of four quantum dots arranged in a square, with two mobile electrons that occupy diagonal positions. Depending on which diagonal they occupy, the cell represents either logic '0' or logic '1'. Neighboring cells interact through Coulombic (electrostatic) forces, allowing information to propagate without any actual current flowing through a conducting path.

This makes QCA extremely attractive for **low-power and ultra-dense logic design**, because it operates based on position and polarization rather than current switching. Furthermore, many reversible logic gates like the Feynman gate, Toffoli gate, and Fredkin gate have been proposed and designed using QCA structures. These QCA-based implementations naturally align with reversible computation principles, reducing energy dissipation and making them suitable candidates for future quantum and nano-scale processors.

In the context of our project, we conceptually combine reversible logic (Feynman gate behavior) with QCA-based design philosophy. The XOR-based embedding and extraction functions used in our Python implementation mimic the logical behavior of a Feynman gate, which can be physically realized using QCA layouts as reported in research. Although our current work is implemented in software using Python and VS Code (i.e., we are not fabricating a real QCA chip), the algorithm and reversible embedding logic are designed with QCA compatibility in mind. This means that, in principle, the same embedding and extraction mechanism could be mapped onto a QCA-based reversible circuit for ultra-low-power, nano-scale hardware realization in the future.

Thus, our project does not only demonstrate **image steganography using reversible logic** at the software level, but it also points towards a **potential hardware implementation using quantum dot cellular automata**. This integration of steganography, reversible gates, and QCA technology

makes the project highly relevant to future generations of nano-electronic and quantum-inspired secure communication systems.

EXISTING MODEL

F5 Steganography Algorithm (Real-Life JPEG Steganography Model)

One of the most widely recognized and practically implemented steganographic models in real-life systems is the F5 Steganography Algorithm. This method was developed by Andreas Westfeld and Andreas Pfitzmann at Technische Universität Dresden, Germany, in the year 1999–2000. Since its introduction, F5 has gained significant attention in both academic and industrial domains due to its ability to embed secret information efficiently within JPEG images. It remains one of the most influential algorithms in the field of digital steganography.

The F5 algorithm has been implemented in real-world applications such as digital watermarking tools, covert image communication systems, forensic and intelligence platforms, and open-source software like StegHide. Cybersecurity and law enforcement agencies have encountered F5-based hidden messages during investigations, demonstrating that this model is not just theoretical but actively used in real digital environments.

Working Principle of the F5 Model

The F5 algorithm is specifically designed for JPEG images and operates in the compressed frequency domain rather than directly on pixel values. The process begins by converting the cover image into JPEG format, where the image is divided into small blocks. Each block undergoes a Discrete Cosine Transform (DCT) to convert spatial-domain pixel values into frequency coefficients. From these coefficients, F5 selects non-zero values for data embedding because modifying them introduces less visible distortion.

A key feature of F5 is matrix encoding, which allows the algorithm to embed multiple bits while altering fewer coefficients, thereby reducing detectability. The embedding itself is performed by

decrementing selected coefficients, a technique known as shrinkage. Once the embedding process is completed, the modified coefficients are recompressed to form the final stego JPEG image. At the receiving end, the process is reversed: the JPEG is decoded, the modified coefficients are examined, and the hidden data is reconstructed. However, due to its irreversible nature, the original cover image cannot be perfectly restored after embedding.

Logic and Operations Used

Although F5 is mathematically oriented, it fundamentally relies on irreversible digital logic. Operations such as bit matching are performed using XOR, embedding checks involve AND and OR logic, coefficient comparisons rely on comparators, and the control flow is managed through counters and registers. Arithmetic units are used for coefficient manipulation. Because these operations are irreversible, the original inputs cannot be reconstructed from the outputs, leading to information loss and increased power dissipation in accordance with Landauer's principle.

Limitations of the Existing Model

Despite its widespread usage and effectiveness, the F5 algorithm has several limitations. It is inherently irreversible, meaning the original cover image cannot be recovered once the embedding is performed. The algorithm requires significant computation involving DCT and coefficient manipulation, leading to higher power consumption and heat generation. Modern steganalysis tools can detect F5-based embedding under certain conditions, reducing its stealth. Additionally, the algorithm is not suitable for nano-scale or low-power hardware platforms and cannot be directly mapped to reversible or quantum logic architectures.

Relevance to the Proposed Project

When compared to the proposed reversible logic–based steganography system, the F5 algorithm highlights clear differences. F5 uses irreversible logic, whereas the proposed model uses reversible XOR/Feynman-inspired behavior. The existing model causes higher power dissipation, while the proposed system conceptually supports lower power usage. F5 does not

allow recovery of the original cover, whereas the reversible model enables perfect retrieval of the secret data. Additionally, F5 is tailored for JPEG images, while the proposed system works at the pixel level in grayscale images and is more suitable for future Quantum Dot Cellular Automata (QCA)–based hardware implementations. Therefore, the F5 algorithm serves as a practical benchmark and demonstrates why a reversible, energy-efficient, and hardware-compatible approach is needed.

PROBLEM STATEMENT

With increasing digital communication, secure transmission of sensitive information is becoming more challenging, as traditional methods either expose the presence of hidden data or suffer from information loss and power inefficiency. Conventional steganography relies on irreversible logic, leading to limited recoverability, higher computational energy, and potential visual distortion. As devices scale toward nano-technology, reducing heat and power dissipation has become critical. Therefore, a secure, imperceptible, and reversible data-hiding technique with future hardware compatibility is urgently required.

SOLUTION

To address these challenges, the proposed solution integrates **steganography with reversible logic principles**, inspired by the behavior of the **Feynman (CNOT) gate**, and validates it through a Python-based implementation. The core concept is to embed secret information in such a way that no logical information is destroyed, ensuring perfect recoverability and improved efficiency. In this system, a grayscale cover image and a secret image are processed pixel-by-pixel. The two most significant bits of each secret pixel are extracted and embedded into the least significant bits of the corresponding cover pixel using a reversible XOR-based operation. This XOR mapping mimics the functionality of the Feynman gate, where the output maintains a one-to-one relationship with the input, enabling the original data to be retrieved without loss.

The embedding process generates a stego image that visually appears unchanged when compared to the cover image, fulfilling the requirement of imperceptibility. During extraction, the same reversible XOR logic is applied using the stego image and original cover image, allowing the secret image to be reconstructed exactly as it was before embedding. This ensures perfect retrieval, which is a major limitation in many traditional irreversible methods. The system is implemented using Python and Visual Studio Code, utilizing libraries such as PIL for image handling, NumPy for efficient pixel-level operations, and scikit-image for quality assessment.

To evaluate performance, quantitative metrics including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) are computed between the cover and stego images. Low MSE, high PSNR, and SSIM values close to 1 confirm that the embedding process introduces minimal distortion, making the stego image virtually indistinguishable to the human eye. These results demonstrate that the reversible approach not only secures the hidden data but also preserves image quality.

Furthermore, the reversible nature of the logic aligns with **Landauer's principle**, which states that information loss leads to energy dissipation. By avoiding irreversible operations, the solution conceptually supports lower power consumption and reduced heat generation. This forms a strong connection to **Quantum Dot Cellular Automata (QCA)** technology, a promising nano-electronic platform that implements logic using electron positioning rather than current flow. Since reversible gates like the Feynman gate have been modeled in QCA structures, the proposed system holds potential for future hardware realization at the nano scale. This could lead to ultra-low-power, compact, and high-speed secure communication devices.

In summary, the solution offers a secure, reversible, low-distortion, and future-ready steganography method. It demonstrates practical effectiveness through software implementation while establishing a foundation for next-generation, energy-efficient hardware based on reversible logic and QCA.

Program Code

Image to Grayscale

```
from PIL import Image

# Cover image

cover = Image.open("cover.png").convert("L")   # grayscale

cover = cover.resize((256, 256))

cover.save("cover1.png")

# Secret image

secret = Image.open("rudra.jpg").convert("1") # binary

secret = secret.resize((64, 64))

secret.save("secret1.png")

print(" Cover and Secret images are ready!")
```

Stegnography code

```
from PIL import Image

import numpy as np

import matplotlib.pyplot as plt

from math import log10, sqrt

from skimage.metrics import structural_similarity as ssim


def feynman_embed(cover_pixel, secret_pixel):
```

```
    secret_bits = (secret_pixel >> 6) & 0b00000011

    cover_cleared = cover_pixel & 0b11111100

    stego_pixel = cover_cleared | ((cover_pixel ^ secret_bits) & 0b00000011)

    return stego_pixel


def feynman_extract(stego_pixel, cover_pixel):

    secret_bits = (stego_pixel ^ cover_pixel) & 0b00000011

    secret_pixel = secret_bits << 6

    return secret_pixel


cover_img = Image.open("cover1.png").convert('L')

secret_img = Image.open("secret.png").convert('L')

secret_img = secret_img.resize(cover_img.size)

cover_array = np.array(cover_img)

secret_array = np.array(secret_img)

stego_array = np.zeros_like(cover_array)

extracted_array = np.zeros_like(cover_array)


for i in range(cover_array.shape[0]):

    for j in range(cover_array.shape[1]):

        stego_array[i, j] = feynman_embed(cover_array[i, j], secret_array[i, j])
```

```
stego_img = Image.fromarray(stego_array.astype(np.uint8))
stego_img.save("stego.png")


for i in range(stego_array.shape[0]):
    for j in range(stego_array.shape[1]):
        extracted_array[i, j] = feynman_extract(stego_array[i, j], cover_array[i, j])


extracted_img = Image.fromarray(extracted_array.astype(np.uint8))
extracted_img.save("extracted.png")


def mse(img1, img2):
    return np.mean((img1.astype(np.float64) - img2.astype(np.float64)) ** 2)
def psnr(img1, img2):
    mse_val = mse(img1, img2)
    if mse_val == 0:
        return 100
    return 20 * log10(255.0 / sqrt(mse_val))
mse_val = mse(cover_array, stego_array)
psnr_val = psnr(cover_array, stego_array)
ssim_val = ssim(cover_array, stego_array, data_range=255)
```

```
print("\nQUALITY EVALUATION RESULTS")

print(f"MSE  (Cover vs Stego): {mse_val:.4f}")

print(f"PSNR (Cover vs Stego): {psnr_val:.2f} dB")

print(f"SSIM (Cover vs Stego): {ssim_val:.4f}")


plt.figure(figsize=(12, 4))


plt.subplot(1, 3, 1)

plt.imshow(cover_img, cmap='gray')

plt.title("Cover Image")

plt.axis('off')


plt.subplot(1, 3, 2)

plt.imshow(stego_img, cmap='gray')

plt.title("Stego Image")

plt.axis('off')

plt.subplot(1, 3, 3)

plt.imshow(extracted_img, cmap='gray')

plt.title("Extracted Secret Image")

plt.axis('off')
```

```
plt.show()

print("\nProcess Completed.")
```

TOOLS USED

The development of this reversible logic–based steganography system relied on several software tools and libraries, each serving a specific role in the implementation, execution, and evaluation of the project. The tools used and their purposes are as follows:

1. Visual Studio Code (VS Code) – This served as the primary development environment for writing, editing, and running the Python program. Its integrated terminal, debugging features, and Python extensions made it efficient for code execution and testing.

2. Python 3.x – The core programming language used to implement the reversible embedding and extraction logic inspired by the Feynman gate. Python enabled pixel-level processing, mathematical operations, and rapid prototyping of the algorithm.

3. Pillow (PIL) – This image processing library was used to load, convert, resize, and save images. It allowed the cover and secret images to be transformed into grayscale format and prepared for pixel manipulation.

4. NumPy – Used for numerical and matrix operations, NumPy converted images into arrays and enabled fast pixel-wise iteration for embedding and extraction. It also supported mathematical computations needed for the algorithm.

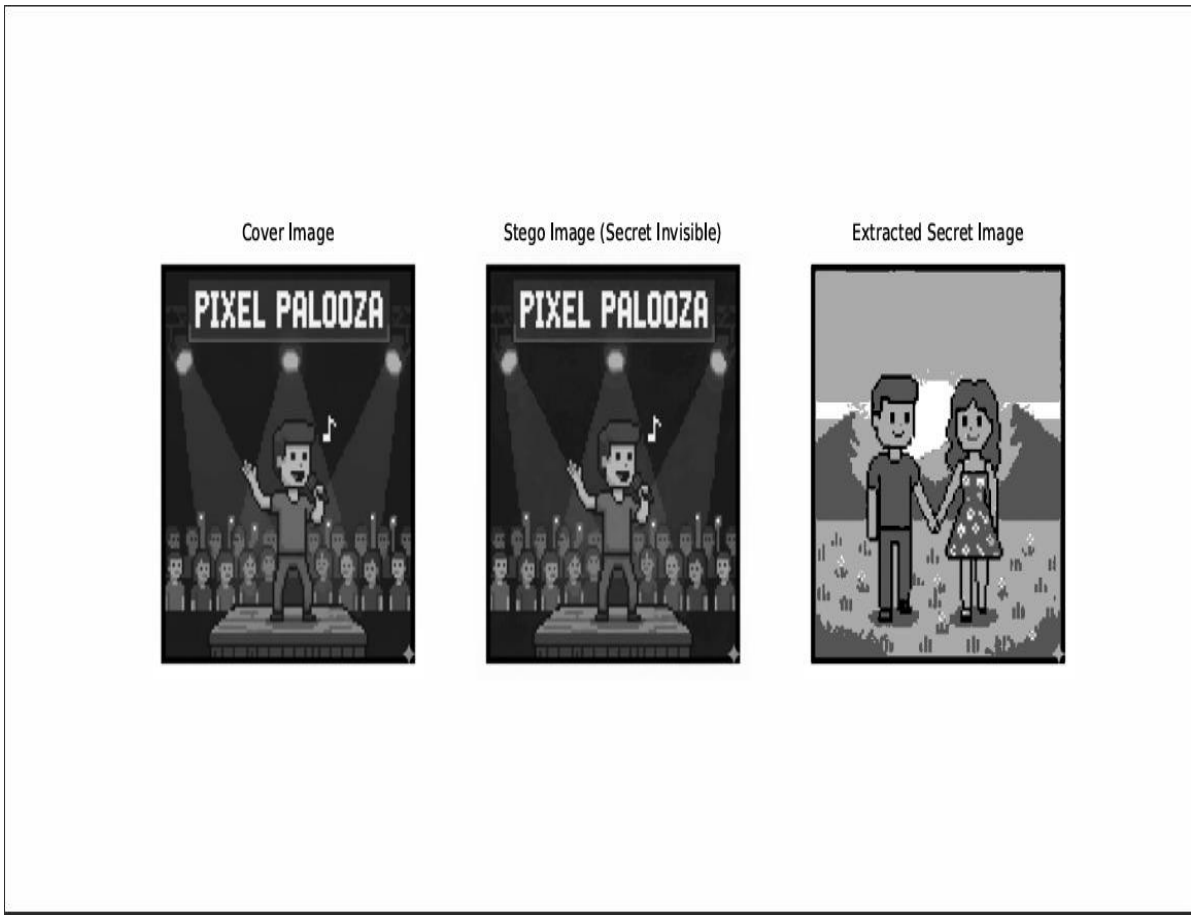
5. Matplotlib – Utilized to display the cover, stego, and extracted images. This helped visually verify that the stego image remained imperceptible and that the secret image was successfully recovered.

6. Scikit-Image (skimage) – Specifically used to calculate the Structural Similarity Index (SSIM), which measures how similar the stego image is to the original cover image. This supported quantitative image quality evaluation.

7. Math Library – The math functions (log10, sqrt) were used to compute the Peak Signal-to-Noise Ratio (PSNR), a key metric for assessing distortion between the cover and stego images.

Together, these tools enabled complete implementation—ranging from image handling and reversible logic execution to visualization and quality assessment—making the project fully functional and scientifically validated in software.

Simulation Results



COMPARISON BETWEEN EXISTING METHOD (F5) AND PROPOSED METHOD

Parameter	**Existing Method – F5 Algorithm**	**Proposed Method – Reversible Feynman Logic (Python/QCA Concept)**
Type of Logic	Irreversible logic	Reversible XOR / Feynman-inspired logic
Domain of Operation	JPEG frequency domain (DCT coefficients)	Pixel-level grayscale domain
Recoverability	Original cover image cannot be restored	Perfect recovery of secret data
Energy / Heat Dissipation	Higher (information loss → energy loss)	Conceptually lower (no logical information loss)
Power Efficiency	Moderate to poor	High theoretical efficiency (reversible computing)
Detectability	Can be detected by modern steganalysis	Very low visual distortion; difficult to detect
Embedding Technique	Matrix encoding + coefficient decrement	XOR-based reversible bit embedding
Visual Distortion	Minimal but noticeable in some cases	Extremely low (high PSNR & SSIM)
Hardware Compatibility	Not suitable for nano/quantum hardware	Future-compatible with QCA & low-power VLSI

Parameter	Existing Method – F5 Algorithm	Proposed Method – Reversible Feynman Logic (Python/QCA Concept)
Bit Loss	Information destroyed during processing	No logical bit loss (reversible mapping)
Computational Complexity	Higher due to DCT and coefficient processing	Lower, simple pixel-wise operations
Implementation	Used in tools like StegHide, forensic cases	Python implementation; hardware possible
Target Media	JPEG images only	Any grayscale image
Suitability for Real-Time Systems	Limited	Highly suitable due to low complexity

ADVANTAGES

The proposed reversible steganography system offers several significant advantages over conventional irreversible methods. One of the most important benefits is **lossless data recovery**, as the reversible XOR-based logic inspired by the Feynman gate enables the secret information to be perfectly extracted without destroying logical information during embedding. This ensures high reliability and accuracy in secure communication. The method also provides **excellent imperceptibility**, since only the least significant bits of the cover image are modified, resulting in very minimal distortion. This is reflected in high PSNR and SSIM values, making the stego image visually almost identical to the original. Additionally, the computational process is **simple and lightweight**, involving basic bitwise operations rather than complex transforms like DCT, which makes the system faster and more suitable for real-time applications. Another key advantage is **lower theoretical power dissipation**, as reversible logic aligns with Landauer's

principle by minimizing information loss, making the concept potentially more energy-efficient. The method is also **future-compatible with Quantum Dot Cellular Automata (QCA) and low-power VLSI hardware**, offering promising potential for nano-scale and quantum-inspired implementation. Furthermore, the system is flexible and easy to implement in software using Python, making it accessible for research, education, and further development.

DISADVANTAGES

Despite its strengths, the proposed system also has certain limitations. One major drawback is that **extraction requires access to the original cover image**, since reversibility depends on the XOR relationship between the cover and stego pixels; without the original cover data, the hidden information cannot be accurately recovered. The embedding capacity is also **limited**, as only two bits per pixel are used, which restricts how much secret data can be embedded in a single image compared to other advanced techniques. While the software implementation is straightforward, a **true hardware realization using reversible logic or QCA is still theoretical**, as practical fabrication tools and commercial reversible hardware are not widely available today. This means that the power-saving advantages are conceptual rather than physically measured. Additionally, the method is sensitive to **image modifications or compression**; if the stego image undergoes cropping, resizing, or JPEG compression, the embedded data may become corrupted. The approach also does not include built-in encryption, so while the data is hidden, it is not encrypted, which may require an additional security layer. Finally, compared to sophisticated steganography algorithms that embed in frequency or transform domains, this pixel-based approach may be more vulnerable to statistical steganalysis in large-scale forensic scenarios.

CONCLUSION

The project successfully demonstrates a reversible logic–based image steganography system inspired by the behavior of the Feynman gate and conceptually aligned with Quantum Dot

Cellular Automata (QCA) principles. By embedding the two most significant bits of the secret image into the least significant bits of the cover image using XOR-based reversible mapping, the system achieves reliable and lossless data extraction. The high PSNR and SSIM values confirm that the stego image remains visually indistinguishable from the cover image, ensuring strong imperceptibility and security. Unlike traditional irreversible methods, this approach preserves logical information, supporting the concept of lower theoretical power dissipation and reduced heat generation as outlined by Landauer's principle. The Python implementation provides a practical and efficient platform for validating the reversible embedding and retrieval process, while also offering a foundation for future hardware realization in low-power nano-scale technologies. Overall, the project establishes an effective, efficient, and future-ready steganographic model that integrates reversible logic, digital image processing, and QCA-oriented design, making it a promising direction for next-generation secure communication systems.

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